Classifying differential equations

Let's start by clarifying the difference between partial derivatives and ordinary derivatives.

Partial derivatives

The **partial derivative** of a function is the derivative of that function with respect to one of the multiple variables in which the function is defined. For instance, given a function f defined in terms of two variables x and y, $f(x, y)$, we know that f has two partial derivatives:

The partial derivative of f with respect to x $\quad \dfrac{\partial f}{\partial x}$

The partial derivative of f with respect to y $\quad \dfrac{\partial f}{\partial y}$

We can't name only one derivative for f, since f is defined in two variables. Instead, we find one partial derivative of f for each of its variables, which is why, for $f(x, y)$, we end up with two partial derivatives.

When we define equations using partial derivatives like these ones, we call them **partial differential equations (PDEs)**. We usually study partial derivatives and their equations in a multivariable/multivariate calculus course, which is often Calculus III or Calculus IV.

That being said, these aren't the kinds of differential equations we'll focus on in this course. We'll spend the vast majority of our time focusing on

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ordinary differential equations, and we'll only touch briefly on partial differential equations at the very end of the course.

Ordinary derivatives

Whereas partial derivatives are indicated with the "partial symbol" ∂, we never see this notation when we're dealing with ordinary derivatives. That's because an **ordinary derivative** is the derivative of a function in a single variable. Because there's only one variable, there's no need to indicate the partial derivative for one variable versus another.

For example, given a function for y in terms of x, which we could write as $y(x)$, its first derivative can be written as $y'(x)$, or as just y', or in **Leibniz notation** as

$$\frac{dy}{dx}$$

So an equation like

$$\frac{dy}{dx} - \sin x \cos x = 2x$$

is an **ordinary differential equation** because it includes the ordinary derivative dy/dx.

Order of the differential equation

The **order** of a differential equation is equivalent to the degree of the highest-degree derivative that appears in the equation. For example, if the

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equation contains only a first derivative, we call it a first order differential equation. Here are some more examples:

Equation	Order
$y' - 2xy = x\cos y$	1
$5y'' - 2y' + 5y = 0$	2
$\frac{d^4y}{dx^4} + e^x\frac{dy}{dx} = 0$	4

In this differential equations course, we'll be focusing primarily on first and second order differential equations. We're starting with first order equations now, and we'll get into second order equations later.

Linear differential equations

When it comes to classifying first order differential equations, we put them into two categories: linear and separable. We'll talk much more about each of these types later. For now, we only want to say that **linear differential equations** are equations given in the form

$$p_n(x)y^{(n)}(x) + p_{n-1}(x)y^{(n-1)}(x) + \ldots + p_1(x)y'(x) + p_0(x)y(x) = q(x)$$

where $p_i(x)$ and $q(x)$ are functions of x. If a first order ordinary differential equation doesn't match this form, we say that it's a non-linear equation.

What we want to take away from this definition of linear equations is that

1. all of the $p_i(x)$ coefficients are functions in terms of only x,

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2. $q(x)$ is also a function in terms of only x,

3. the function y is never defined to a higher power than 1 (we should only see y and its derivatives y', y'', y''', etc., never something y^2, $\sin y$, e^y, etc.

Let's add a linear/non-linear classification to our table from earlier.

Equation	Order	Linearity
$y' - 2xy = x\cos y$	1	Non-linear
$5y'' - 2y' + 5y = 0$	2	Linear
$\frac{d^4y}{dx^4} + e^x\frac{dy}{dx} = 0$	4	Linear

The first equation in this table is non-linear because $q(x) = x\cos y$, which means $q(x)$ is a function defined in terms of both x and y, not just x alone.

These second and third equations are linear equations because they meet the three conditions we outlined. Notice that, in both linear equations, $q(x) = 0$. When this is the case, we say that the linear equation is **homogeneous**. As you might suspect, when $q(x) \neq 0$ we call the linear equation **non-homogeneous**.

Let's work through some more examples so that we get comfortable classifying differential equations.

Example

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Identify the order and linearity of each differential equation, then say whether or not each linear equation is homogeneous.

1. $8y''' + 2y' + \cos y = e^x$
2. $2y'' + 5y' = xy$
3. $e^x y' - e^{x+y} = e^{2x}$
4. $y''' + 5y' = y \cos x$
5. $y'' - 3y = \sin x$
6. $xy' + 2y = x^2 - x + 1$

1. The equation $8y''' + 2y' + \cos y = e^x$ contains a third derivative, so it's a third order equation. Because the equation contains $\cos y$, it's non-linear.

2. The equation $2y'' + 5y' = xy$ contains a second derivative, so it's a second order equation. We can rewrite it as $2y'' + 5y' - xy = 0$, which allows us to see that the equation is linear and homogeneous.

3. The equation $e^x y' - e^{x+y} = e^{2x}$ contains a first derivative, so it's a first order equation. Because the equation contains e^y, it's non-linear.

4. The equation $y''' + 5y' = y \cos x$ contains a third derivative, so it's a third order equation. We can rewrite it as $y''' + 5y' - y \cos x = 0$, which allows us to see that the equation is linear and homogeneous.

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5. The equation $y'' - 3y = \sin x$ contains a second derivative, so it's a second order equation. The equation is already in the form of a linear equation, but $q(x) = \sin x \neq 0$, so it's non-homogeneous.

6. The equation $xy' + 2y = x^2 - x + 1$ contains a first derivative, so it's a first order equation. The equation is already in the form of a linear equation, but $q(x) = x^2 - x + 1 \neq 0$, so it's non-homogeneous.

We can summarize our findings in a table.

Equation	Order	Linearity	Homogeneity
$8y''' + 2y' + \cos y = e^x$	3	Non-linear	
$2y'' + 5y' = xy$	2	Linear	Homogeneous
$e^x y' - e^{x+y} = e^{2x}$	1	Non-linear	
$y''' + 5y' = y \cos x$	3	Linear	Homogeneous
$y'' - 3y = \sin x$	2	Linear	Non-homogeneous
$xy' + 2y = x^2 - x + 1$	1	Linear	Non-homogeneous

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